PINNs for Liquid-Solid Systems with Phase Change

B.-E. Madir, F. Luddens, C. Lothodé, I. Danaila

Contact: bahae-eddine.madir@univ-rouen.fr

Laboratoire de Mathématiques Raphaël Salem, Université de Rouen Normandie

Introduction

Phase Change Materials (PCMs) are widely used in industrial applications, particularly in building insulation, energy storage systems, and thermal management devices in electronics. PCMs involve liquid–solid phase transitions, which can be modeled using **the Stefan problem**, when **the natural convection** in the liquid phase is neglected. The complete model for PCMs is based on the Navier–Stokes–Boussinesq equations coupled with an enthalpy equation.

We explore the use of Physics-Informed Neural Networks (PINNs) [RPK19] to solve liquid–solid systems with phase change. We developed a computational framework that can be extended to data assimilation and inverse problems for PCM optimization.

Stefan's Problem

When heat transfer in a PCM is dominated by *conduction*, the temperature θ in each phase (liquid and solid) can be described by a diffusion equation (1). At the interface between the two phases ($x = s(t)$), energy conservation must be enforced through the Stefan condition (2):

$$\frac{\partial \theta_j}{\partial t} - \mathrm{Fo}\,\frac{\partial^2 \theta_j}{\partial x^2} = 0, \quad j \in \{1, 2\}, \tag{1}$$

$$\frac{\partial \theta_2}{\partial n} - \frac{\partial \theta_1}{\partial n} = \mathrm{Fo}^{-1}\,\mathrm{Ste}^{-1}\,s'(t), \quad x = s(t). \tag{2}$$

Where Fo and Ste are dimensionless numbers.

Natural Convection

In the liquid phase, the presence of a significant temperature difference generates a buoyancy force $\vec{F}$, causing warmer (less dense) fluid to rise and colder (denser) fluid to sink. Natural convection is governed by the Navier–Stokes–Boussinesq equations:

$$\begin{aligned} \nabla \cdot \boldsymbol{u} &= 0, \\ \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} + \nabla p - \frac{1}{\mathrm{Re}}\nabla^2 \boldsymbol{u} - \vec{F}(\theta) &= 0, \\ \frac{\partial \theta}{\partial t} + (\boldsymbol{u} \cdot \nabla)\theta - \frac{1}{\mathrm{Re}\,\mathrm{Pr}}\nabla^2 \theta &= 0. \end{aligned} \tag{3}$$

Where Re and Pr are dimensionless numbers.

Physics-Informed Neural Networks (PINNs)

Neural networks are universal approximators [Cyb89]. Therefore, the solution of (1)–(2) (or (3)) can be approximated using a neural network. This is achieved by transforming the PDEs into residuals $\mathcal{R}^{(k)}$ and defining the associated loss function $\mathcal{L}_r$, with all required derivatives computed efficiently via Automatic Differentiation (AD). Similarly, losses are defined for the initial condition $\mathcal{L}_i$ and boundary conditions $\mathcal{L}_b$.

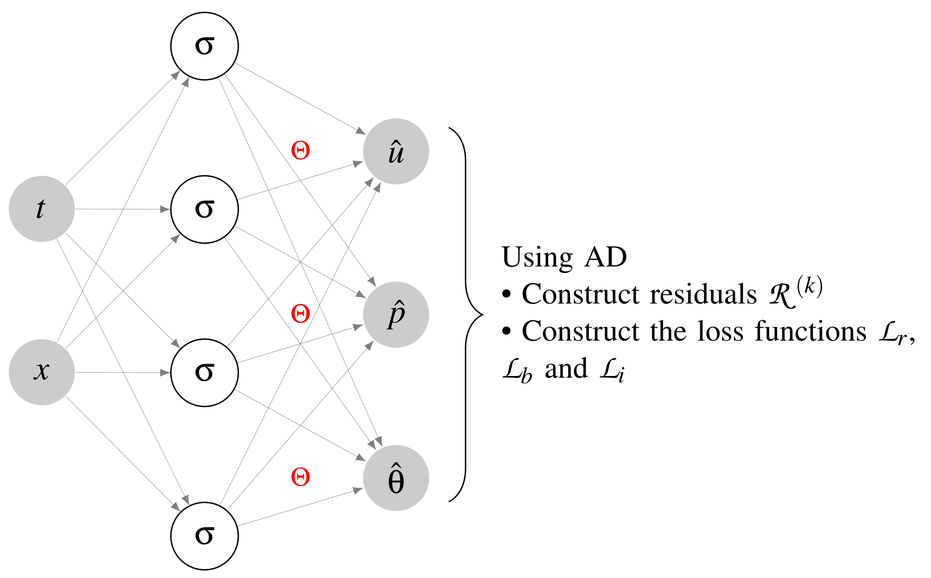
The total loss is defined as the *weighted* sum:

$$\mathcal{L}(\Theta) = \omega_i\,\mathcal{L}_i + \omega_b\,\mathcal{L}_b + \omega_r\,\mathcal{L}_r. \tag{4}$$

The network parameters are updated according to gradient descent:

$$\Theta^{j+1} = \Theta^j + \eta^j\,\mathbf{D}^{(j)}, \tag{5}$$

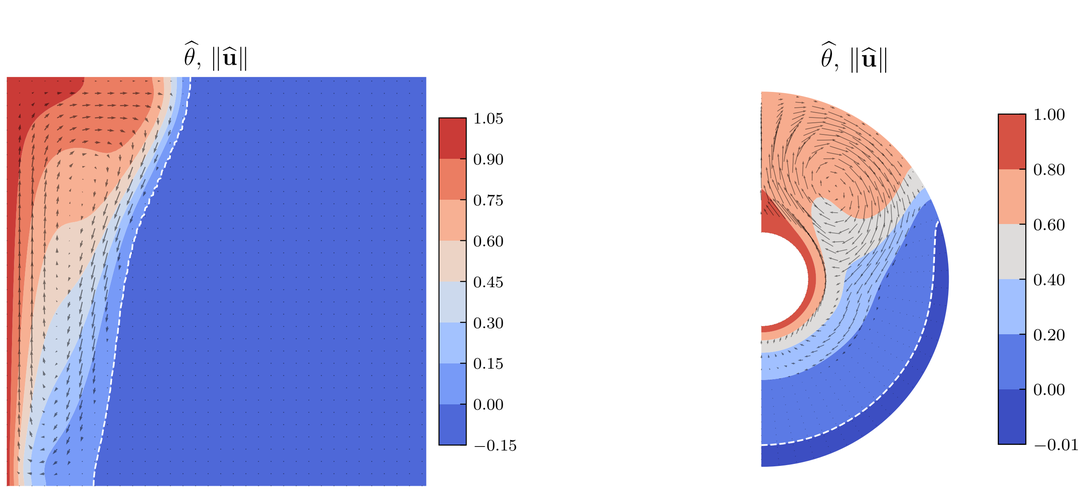
where η^j is the learning rate and $\mathbf{D}^{(j)}$ is the update direction, which depends on the gradients $\nabla_\Theta \mathcal{L}(\Theta^l)$ for $l \leq j$.



PINNs for PCMs

For the Stefan problem (1) – (2), we proved that PINNs can provide accurate approximations for different phase-change regimes. We employed various strategies, such as dynamic weighting approaches, to enhance the performance of PINNs. Different configurations were investigated, including two-dimensional cases for both melting and solidification [Mad+25].

We successfully extended our work to natural convection (3), and subsequently to PCM melting. The results were validated against high-fidelity numerical simulations (finite elements) and experimental data.



Melting of a PCM, simulated with PINNs (the liquid–solid interface is highlighted in white).

Perspectives

In practical situations, the goal is to optimize the performance of a PCM used as a heat managment or storage device (fusion temperature, absorbed latent heat, etc) as a function of a set of parameters $\boldsymbol{\alpha}$, which may include the geometry and shape of the PCM, the position of fins, or the type of PCM itself. In such cases, the neural network can be structured to take $\boldsymbol{\alpha}$ as an input variable and then used as a differentiable surrogate model. This enables efficient optimization strategies without requiring repeated full-scale simulations.

References

[Cyb89] George Cybenko. "Approximation by superpositions of a sigmoidal function". In: *Mathematics of control, signals and systems* 2.4 (1989), pp. 303–314.

[Mad+25] Bahae-Eddine Madir, Francky Luddens, Corentin Lothodé, and Ionut Danaila. "Physics Informed Neural Networks for heat conduction with phase change". In: *International Journal of Heat and Mass Transfer* 252 (2025), p. 127430.

[RPK19] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations". In: *Journal of Computational physics* 378 (2019), pp. 686–707.